Team RoboRevolutionaries

- ## Problem Statement

  Actual Problem Statement

  Summary: The Flipkart Grid 5.0 competition tackles e-commerce logistics challenges by focusing on "Automated Singulation." Teams of up to six members must create a system capable of detecting, picking, and placing individual packages in a designated drop zone, emphasising proper barcode orientation. The timeline includes registration, submission rounds, and an in-person final event. Round I involves handling three package types based on their sizes, judging criteria for successful movement and specific label conditions. Teams submit pictures, a bill of materials, and videos showcasing their system's performance for evaluation. The competition aims to address a crucial aspect of e-commerce logistics through innovative automation solutions.

- ## Team Members

  Note: All team members are from the Indian Institute of Technology(BHU), Varanasi, with 2026 as the Year of Graduation.

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- ## Videos

  - **Explanation Video:** https://youtu.be/PPaXgys5mMI?si=JBkboL0zyfsZSfKN
  - **Final Submission Video:** https://youtu.be/LkgdX7k3M0I?si=prm898AWdBYGlQn_
  - **Image Processing Video:** https://youtu.be/a30uo8gdDdI?si=NH9oLw4nBpMkpdaG

- ## Images of our Setup



⬅️Picture of the entire setup

The Core XY Gantry System ➡️ Along with the connections made.





⬅️ **Robotic Arm with inverse kinematics And packages to be moved to the Drop zone.**

- ## Result

  The result of the competition was that our Team got the remark, “Great Attempt”, and our team was ranked 30th, but unfortunately, only 22 teams were selected for the finals.
  Result:
  https://docs.google.com/spreadsheets/d/1Ftd7ouZ43M6Hx2HPeq_U_D1-9SpdA2jmmnkpaKFJqxg/edit#gid=0

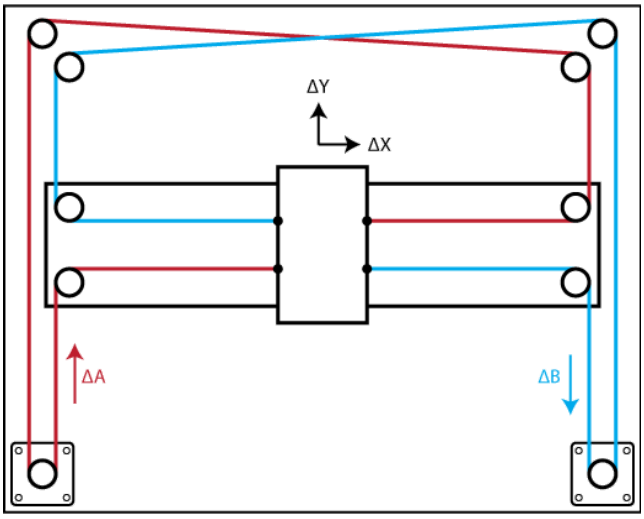
- ## Materials Required

  1. Arduino Mega Board
  2. Stepper Motors (x2)
  3. DC motor
  4. Vacuum Pump

5. Rubber belt
6. Grooves
7. Screws and bolts
8. Iron Rods
9. Tables (For support)
10. Plastic Tube and funnel
11. Wood
12. M-seal
13. PVC pipes

• Working Principle

1. **Core X-Y:** The robotic arm moves in XY coordinates with the help of a Core XY gantry system. The Core XY gantry system works with two stepper motors and a belt system, as described below in the picture. It is also the fundamental principle of 3D printers and CNC machines.

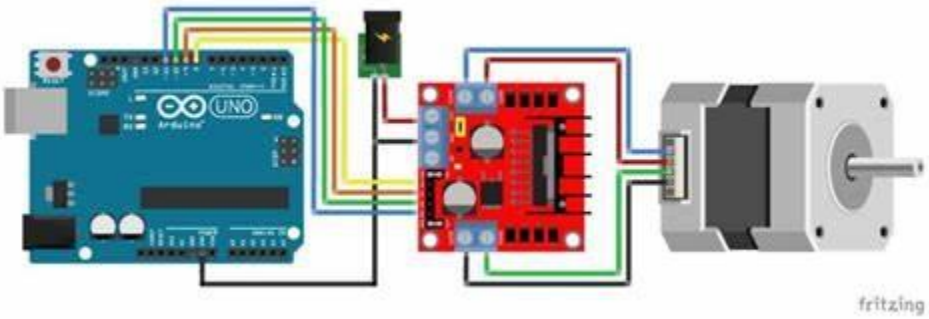


Equations of Motion:

$\Delta X = {}^{1}/_{2}(\Delta A + \Delta B), \quad \Delta Y = {}^{1}/_{2}(\Delta A - \Delta B)$

$\Delta A = \Delta X + \Delta Y, \quad \Delta B = \Delta X - \Delta Y$

2. **Connections with Stepper Motor:** Stepper motors are used due to their increased accuracy. Each stepper Motor is connected to a L298 motor Driver with the following circuit diagram.



3. **The Robotic Arm:** The robotic arm has three joints that use Inverse Kinematics to adjust its height concerning the package. The topmost joint rotates along with the DC motor attached to the gantry. The lowermost joint, restricted to only Up and Down movements through the PVC pipe, moves up and down with every 180-degree DC motor rotation. The middle joint is placed such that the resting position of the entire arm is automatically adjusted if the package is at a greater height (also explained in the video).

4. **The suction:** The suction is created using a 15L air vacuum pump connected to a tube connected to a plastic funnel. As the air gets sucked out of the tube with the help of a pump, the funnel picks up the package due to the vacuum created in the funnel and the package.

5. **Connections with DC motor and vacuum pump:** Both the DC motor connected to the robotic arm and the vacuum pump are connected to the Arduino Board with the help of L298 Motor Drivers through the following circuit diagram.

